

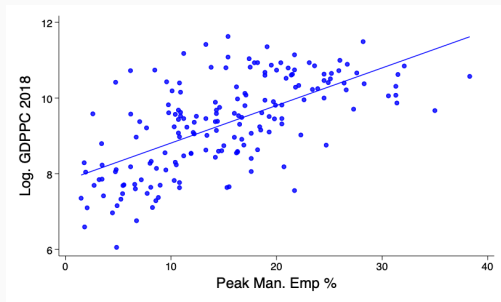
From Agriculture to Services: Development without Industrialization

Tancredi Rapone¹

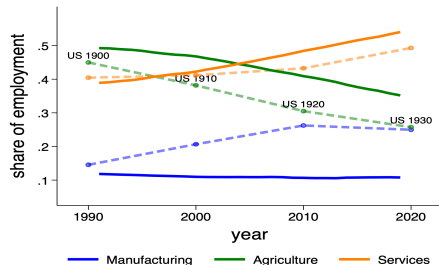
SAEe, December 2025

¹Barcelona School of Economics (BSE) and Universitat Autònoma de Barcelona (UAB)

Motivation



(a) Manufacturing and GDPPC



(b) Emp. Shares over Time

Figure 1: Scatterplot of 2019 GDPPC against peak employment share in manufacturing (left) and sector shares in 42 developing countries (right). Data Sources: Maddison (2020) and ILOEST-Gapminder (2024).

► Table

Q: Can developing countries close the income gap without manufacturing? Should governments implement industrial policy to foster industrial expansion?

Roadmap:

- Document three stylized facts:
 - ⇒ Countries with higher manufacturing peaks $\Rightarrow$ faster growth, better education, and quicker demographic transitions.
- Build a unified growth model to understand the implications of bypassing manufacturing on future growth possibilities.
- Calibrate to the US and change a preference shifter to simulate transitions with different peak manufacturing emp. shares.

Literature

Literature Review

I am contributing to several literatures:

- **Structural Change and Development:** Huneus and Rogerson (2023); Fujiwara and Matsuyama (2024); Sposi, Yi and Zhang (2021); Buera, Kaboski, Mestieri and O'Connell (2024), Kruse, Mensah and de Vries (2022); Rodrik (2016); Duarte and Restuccia (2010); Syrquin Chenery (1989); Kuznets (1959).
- ⇒ Add UGT to understand the consequences of premature de-industrialization on cross country income inequality.
- **Industrial Policy:** Juhász, Lane and Rodrik (2023); Chang and Andreoni (2020); Aghion, Cai, Dewatripont, Du, Harrison and Legros (2015); Rodrik (2008); Herrendorf, Rogerson and Valentinyi (2022).
- ⇒ Integrate structural change in a growth model.
- **UGT:** Galor and Weil (2000); Voigtlander and Voth (2013); Doepke (2004); Hansen and Prescott (2002); Parente and Prescott (2004).
- ⇒ Incorporate sectoral differences in models of UGT.

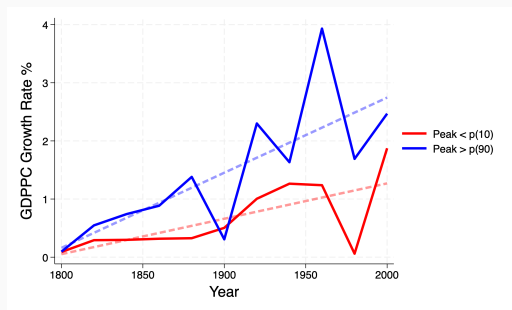
Data Sources

Data Sources

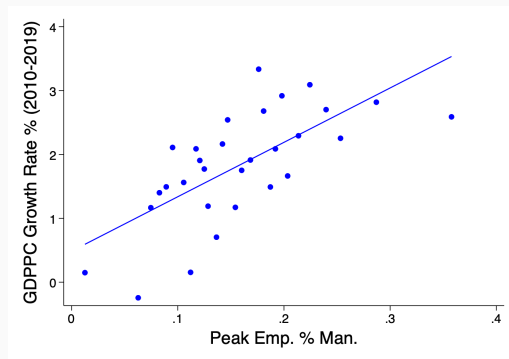
- **Sectoral employment shares:**
 - **Groningen Growth and Development Center (GGDC):**
 - Economic Transformation Database covers 41 low and middle income countries over 1990-2018.
 - 10 Sector Database covers 33 middle/high/low income countries over 1960 - 2010.
 - **ILO via Gapminder:** Remaining countries not covered by GGDC are filled in using Gapminder's data on sectoral employment shares.
 - **US Census of Population:** microdata covering the universe of population with information on employment, occupation and industry at the county level over 1850-present.
- **Fertility:**
 - **IPUMS International:** microdata on fertility and employment sector for 211 country/year samples.
 - **UN via Gapminder:** aggregate fertility rates 190 countries 1800-present.
- **Education:**
 - **Barro and Lee:** 2021 update provides data on years of schooling and human capital quality for 80 high-low and middle income countries over 1960-2020.

Stylized Facts

Stylized Fact 1: GDP p.c. Growth Rates and Manufacturing



(a) Over time



(b) Across countries

Figure 2: (left) Average GDP per capita growth rate over 1800-2020 grouped by top 10% peak emp. share in manufacturing and bottom 10%, for this figure I remove major oil exporters; (right) scatterplot of average GDP per capita growth rates over 2010-2020 against the peak employment share of manufacturing. Data source: Maddison and Gapminder.

Stylized Fact 2: Manufacturing and Education

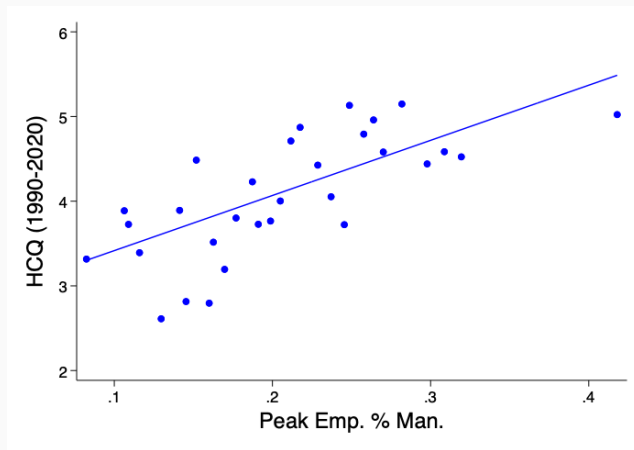
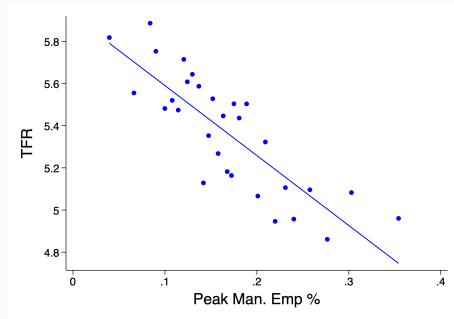
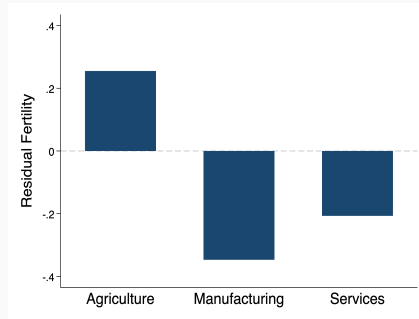


Figure 3: Binned scatterplot of the average Human Capital Quality (HCQ) against the peak manufacturing employment share. Data source: Gapminder and Barro Lee (2015).

Stylized Fact 3: Fertility and Structural Change



(a) Across countries



(b) Within countries

Figure 4: (left) binned scatterplot of average total fertility rates (TFR) against the peak manufacturing share of employment in 175 countries over 1800–2018 controlling for initial fertility and initial GDP per capita; (right) average residual fertility across sectors within country-year samples of childbearing aged women. Data source: Maddison (2023) and Gapminder (left), IPUMS International (right).

Model

The model is an extension of Galor and Weil (2000) and Lagerlof (2006).

Households:

- **New** $\Rightarrow$ Heterogeneous $i \in \{a, m, s\}$ and live for two periods
- Receive education when young and make labor supply, consumption, fertility and education decisions when old
- QQ tradeoff: the effect of education on human capital depends on the rate of technological progress

Production and dynamics:

- **New** $\Rightarrow$ Representative firms produce agricultural, manufacturing and services goods in a competitive market
- **New** $\Rightarrow$ Technological progress depends on population and government provision of education ρ_t but faces *sector-specific congestion effects* (following Jones, 1995)

Household Problem

Each period households in sector $i \in \{a, m, s\}$ solve the following problem:

$$\tilde{U}_i = \max_{c_{at}, c_{mt}, c_{st}, n_{it}, e_{it}} (1 - \gamma) \ln(C_t - \bar{c}) + \gamma \ln(n_{it} h_{i,t+1}) + \eta_i \quad (1)$$

such that:

$$\begin{aligned} h_{i,t+1} &= \frac{e_{it} + \rho_t \tau_i}{e_{it} + \rho_t \tau_i + g_t} \\ p_{at} c_{at} + c_{mt} + p_{st} c_{st} &\leq (1 - n_{it}(e_{it} + \tau_i)) w_t \\ \sum_{j \in \{a, m, s\}} \omega_j^{\frac{1}{\sigma}} \left(\frac{C_{jt}}{C_t^{\varepsilon_j}} \right)^{\frac{\sigma-1}{\sigma}} &= 1 \end{aligned}$$

With $\gamma, \tau \in (0, 1)$, $\sigma \in (0, 1)$, and p_m normalized to 1 and $\eta_i, \eta_j \in \mathbb{R}^2$ s.t. $\tilde{U}_i = \tilde{U}_j \quad \forall \quad i, j$.

Each sector produces using labor and capital with the following CRS technology:

$$Y_{it} = p_{it} A_{it} K_{it}^{1-\nu} (h_t L_{it})^\nu \quad (2)$$

Where capital is obtained on a world capital market with a fixed interest rate $\bar{r}$ implying:

$$Y_{it} \equiv p_{it} A_{it} (h_t L_{it}) \left(\frac{(1-\nu) p_{it} A_{it}}{\bar{r}} \right)^{\frac{1-\nu}{\nu}}.$$

With $\nu \in (0, 1)$.

Dynamics of Population and Technology

Define $\bar{x} \equiv \sum_i s_i x_i$. Population evolves according to fertility decisions:

$$L_{t+1} = \bar{n}_t L_t \quad (3)$$

Technology evolves in each sector according to:

$$\frac{A_{i,t+1}}{A_{i,t}} = (1 + \underbrace{(\bar{e}_t + \rho_t \bar{r}_t) L_t^{\phi_i}}_{g_{i,t}}) \quad (4)$$

Where:

$$\rho_t = v \left(\frac{\max \{e_\tau\}_{\tau < t}}{b} \right)^\xi$$

Balanced Growth Path and Structural Change

There exists an asymptotic **Balanced Growth Path (BGP)** such that:

1. The employment share of services converges to one:

$$\lim_{t \rightarrow \infty} s_{s,t} = 1,$$

while fertility converges to its replacement level:

$$\lim_{t \rightarrow \infty} n_t = 1.$$

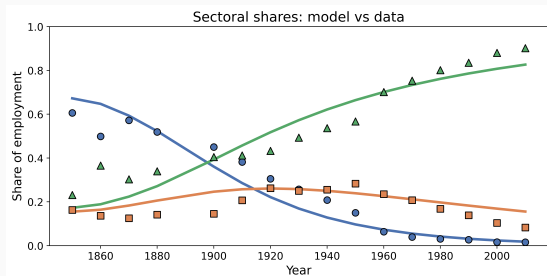
2. Along the BGP, education, growth, and population are constant at:

$$e^* = \gamma - \tau, \quad g^* = \frac{[\gamma + (\rho - 1)\tau]^2}{(1 - \rho)\tau}, \quad L^* = \left[\frac{\gamma + (\rho - 1)\tau}{(1 - \rho)\tau} \right]^{1/\phi_s}.$$

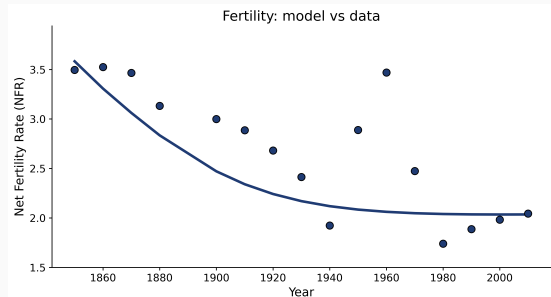
3. The long-run growth rate g^* is **increasing in** ρ .

Calibration

Calibration: USA 1850-2020



Structural Change



Demographic Transition

► GDPPC

► Productivity

► Sectoral Fertility

Table 1: Calibrated parameters and targeted moments

Parameter	Description	Value	Target moment	Data	Model
w_a	Utility weight	1.852	Mean agriculture emp. share	0.259	0.246
w_s	Utility weight	5.492	Mean services emp. share	0.560	0.544
ϵ_a	IE of agriculture demand	0.025	Agriculture emp. share (1850)	0.606	0.672
ϵ_s	IE of services demand	1.161	Services emp. share (1850)	0.231	0.173
τ	Child time cost	0.057	Fertility (1850)	3.496	3.583
v	Education spillover	0.509	Fertility (average over path)	2.688	2.404
ξ	Education spillover	0.941	Fertility (1930)	2.413	2.170
b	Education spillover	0.607	GDPpc growth (1850–1930)	2.932	3.328
γ	Utility weight	0.083	GDPpc growth (1850–2020)	14.471	14.032
L_0	Initial population size	18.767	Manufacturing peak year	1950	1920
ϕ_m	Manufacturing productivity	0.729	Peak manufacturing emp. share	0.283	0.261
ϕ_a	Agriculture productivity	0.755	g_a/g_m	1.194	1.090
ϕ_s	Services productivity	0.506	g_s/g_m	0.452	0.445
τ_a	Agriculture child time cost	0.473	Mean agriculture fertility (last 50)	3.292	3.604
τ_s	Services child time cost	0.984	Mean services fertility (last 50)	2.000	1.995

► Fixed parameters

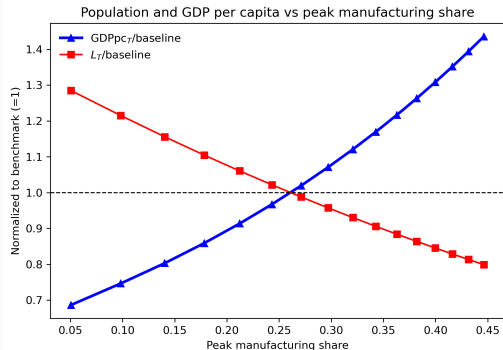
Results

Questions we can ask the model:

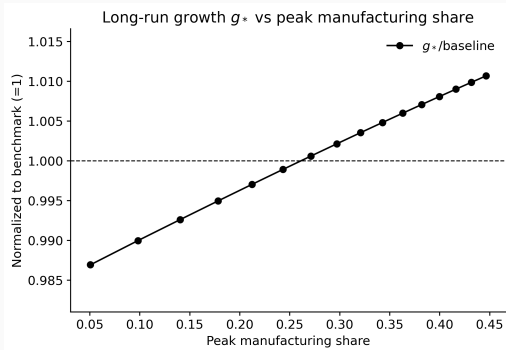
- Can we reduce the US' peak manufacturing employment share? What happens to its growth rate, population and GDPPC today?
 - ⇒ Change ω_m or add a wedge to the relative price of services $p_s^{Eff} = (1 + \tau_{ind})A_m/A_s$.
- How does structural change affect the demographic transition ?
- How does the model account for the cross-country facts we observe in the data? These are
 - Negative relationship between a country's peak manufacturing employment share and its growth rate.
 - Higher population growth in economies which do not go through manufacturing.
 - Higher levels of education quality (ρ) in countries with higher peaks of manufacturing activity

US Results

Results: GDPPC and Population



(a) $L_T, GDPPC_T$



(b) g^*

Figure 5: Final GDPPC and Population for economies with different peak employment shares in manufacturing.

Results: Decomposing the Demographic Transition

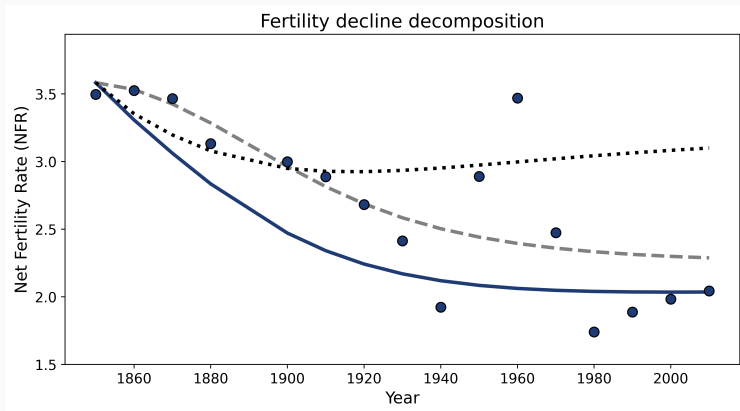


Figure 6: Decomposition of the fertility decline. Dots are data points. Solid line is the (targeted) model path of average fertility. The grey dashed line keeps education investments at their level in 1850. The dashed blue line shows the decline in fertility keeping the sectoral employment shares constant at their level in 1850.

Cross-Country Results

Validation: HCQ vs ρ

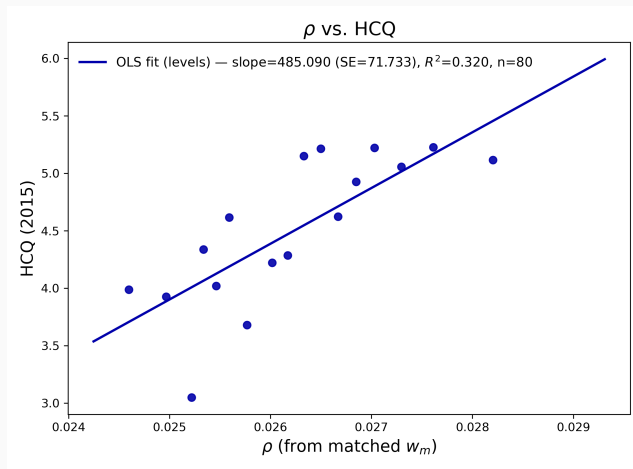
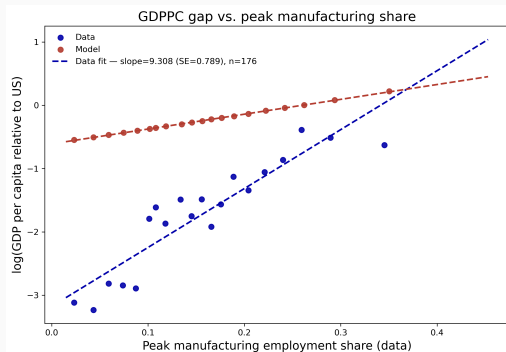
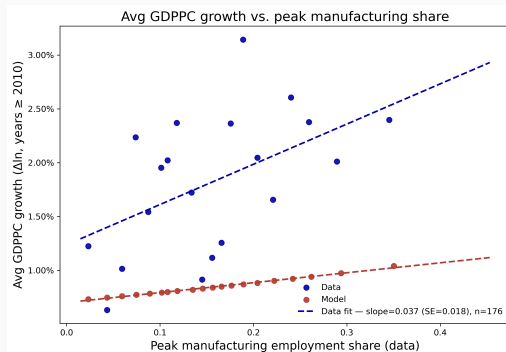


Figure 7: Binscatter plot of endogenous model parameter ρ against data HCQ for economies with different peak employment shares in manufacturing.

Cross Country Results: GDPPC Gaps and Growth Rates



(a) $\log(GDPPC_{c,T}/GDPPC_{USA,T})$



(b) $g_{c,T}^*, g_{c,T}^T$

Figure 8: Final GDPPC and Population for economies with different peak employment shares in manufacturing.

Cross Country Results: Education and Population Growth Factors

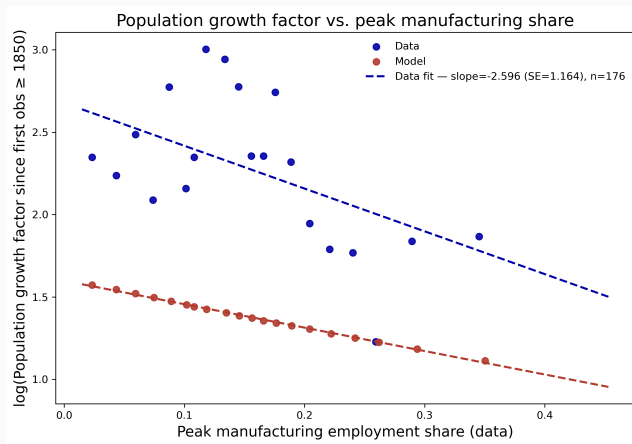


Figure 9: Binscatter plot of $\ln(Pop_{2020}/Pop_{1850})$ for economies with different peak employment shares in manufacturing.

Main idea: A growth model with structural change and endogenous fertility explains key global patterns.

- ⇒ Countries with higher manufacturing peaks $\Rightarrow$ faster growth, better education, and quicker demographic transitions.

Model exercise: Calibrate to the US (1850–2010); vary a preference shifter to simulate economies with more/less manufacturing.

Results: The model explains:

- 23.8% of GDP per capita **level gaps**
- 27.2% of GDP per capita **growth gaps**
- 53.9% of population **growth gaps**

Policy

Policy Experiments:

- Trade protection as a tool to protect industry:
 - Armington setup, small open economy (no effect on world prices). Home competes with ROW which is identical to it except having a Δ_t year head start.
 - Government implements at date T_{pol} a permanent trade openness reform, lowering its tariff from ∞ to τ^o .
- Education and fertility subsidies and taxes:
 - Government implements at date T_{pol} a proportional subsidy (or tax) to education and fertility choices which is financed through a lump sum transfer:

$$\tilde{C}_i = w_t(1 - n_i(e_i + \tau_i)) + s_e e_i + s_n n_i + B_t$$

- Industrial policy (WIP):
 - Government introduces a wedge to subsidize manufacturing employment and balances its budget by taxing services.

Trade Policy

Trade extension I: Tradable prices and demand

Tradability. Agriculture (a) and manufacturing (m) are tradable; services (s) are non-tradable.

Armington aggregation for tradables (Home). For each $i \in \{a, m\}$, Home consumes a CES composite of a Home and a foreign variety:

$$p_{i,t}^{\text{Arm}} = \left[(1 - \lambda_i)(p_{i,t}^H)^{1-\theta} + \lambda_i(\tau_o p_{i,t}^F)^{1-\theta} \right]^{\frac{1}{1-\theta}}, \quad \theta > 1.$$

$\tau_t \geq 1$ is an iceberg trade cost and $p_{i,t}^F$ is the foreign (ROW) price. Services remain non-tradable:
 $p_{s,t}^{\text{Arm}} = p_{s,t}^H$.

Trade policy. An openness instrument $\phi_o \in [0, 1]$ maps into iceberg costs τ_o :

$$\phi_t = 1 \Rightarrow \text{autarky}, \quad \phi_t = 0 \Rightarrow \text{free trade}, \quad \tau_o = \frac{1}{1 - \phi_o}$$

with liberalization happening at T_{trade} .

Trade extension II: Employment shares and dynamics

Domestic absorption under Armington. For each tradable $i \in \{a, m\}$, the domestic expenditure share is

$$\alpha_{i,t}^H = (1 - \lambda_i) \left(\frac{p_{i,t}^H}{p_{i,t}^{\text{Arm}}} \right)^{1-\theta}.$$

From consumption to employment shares. Domestic production/employment shares are:

$$\tilde{s}_{a,t}^L = s_{a,t}^C \alpha_{a,t}^H, \quad \tilde{s}_{m,t}^L = s_{m,t}^C \alpha_{m,t}^H, \quad \tilde{s}_{s,t}^L = s_{s,t}^C, \quad s_t^L = \frac{\tilde{s}_t^L}{\sum_j \tilde{s}_{j,t}^L}.$$

Key modeling choices.

- **Small open economy:** Home takes foreign prices $p_{i,t}^F$ as exogenous (from an autarkic ROW path).
- **Balanced trade:** In each period, total import value equals total export value; import absorption is proportionally scaled (real exchange rate adjustment) to satisfy $\sum_i M_{i,t} = \sum_i X_{i,t}$.

Trade Openness: Effects on Employment Shares

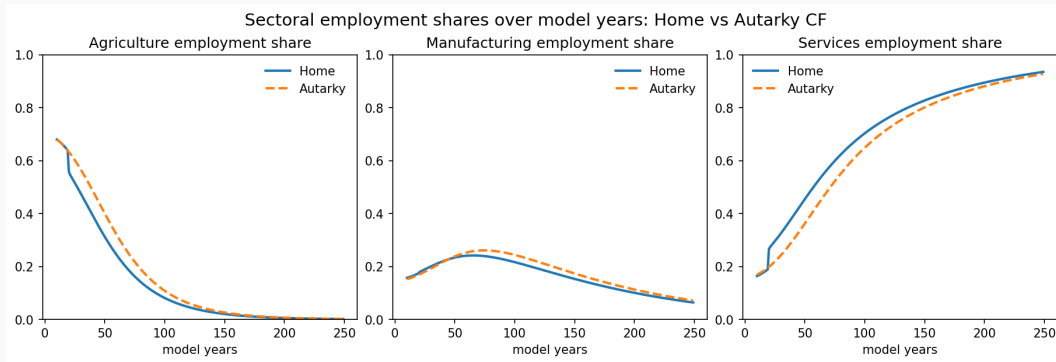
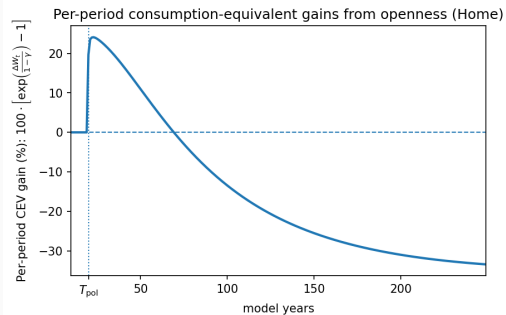
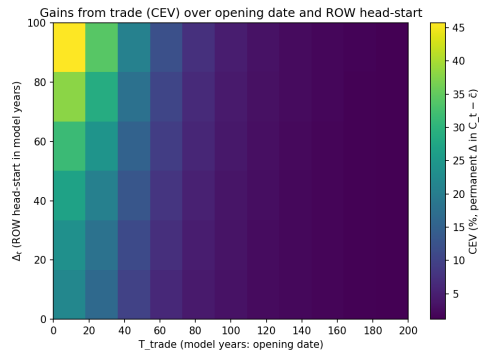


Figure 10: Trade liberalization effects. At $T_{pol} = 50$ the government permanently opens the economy to trade with a tariff of 25%. Trade openness exposes the economy to trade with ROW which is 50 years ahead of the home economy.

Trade Openness: Welfare Gains



(a) Gains from trade over time



(b) CEV

Figure 11: Model simulations and gains from trade. Left panel shows the percentage difference of welfare for the open economy relative to the autarky counterfactual over time. Right panel shows CEV computed using a discount rate of 2% for different openness scenarios varying the head-start of the ROW and the opening date.

Trade Openness: CEV, Growth and Manufacturing

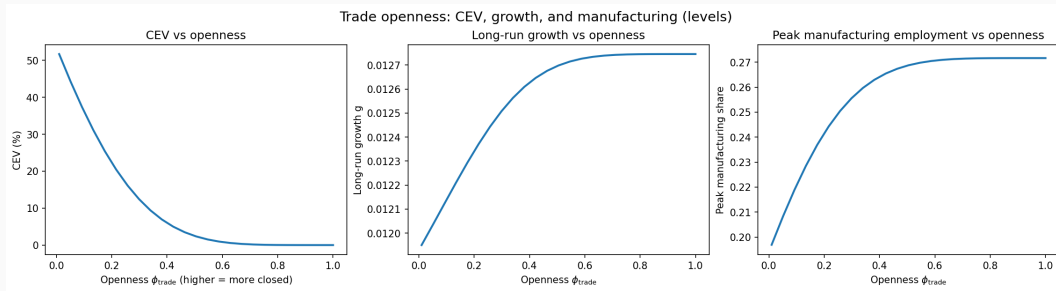


Figure 12: Trade liberalization effects. From left to right, effects on consumption equivalent variation (CEV), long run growth rate and the peak manufacturing employment share.

Trade Openness: Comparative Advantage

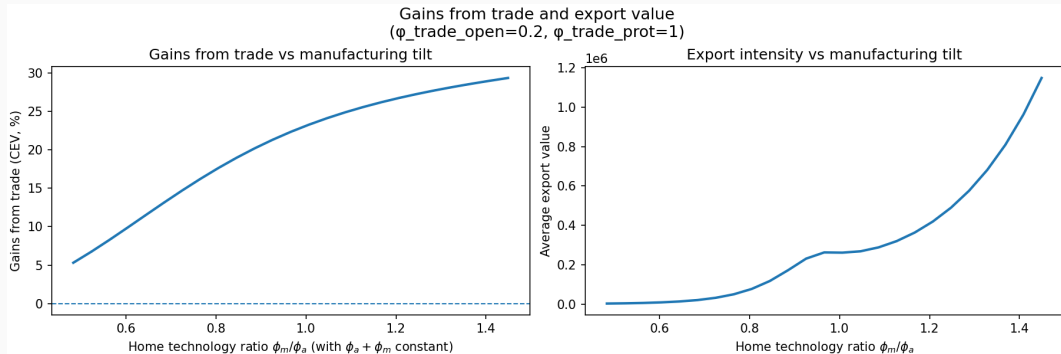


Figure 13: Trade liberalization effects for economies with different ϕ_m/ϕ_a ratios (home ϕ_m/ϕ_a is fixed at calibration values).

► CEV Matrix

Education and Fertility Policy

Setup: Implementation in the Household Problem

Household budget constraint with subsidies:

$$\tilde{C}_i = w_t(1 - n_i(e_i + \tau_i)) + s_e e_i + s_n n_i,$$

where s_e is an education subsidy (or tax) and s_n a fertility subsidy (or tax).

Government budget: in every period the government balances its budget through a lump-sum tax B_t :

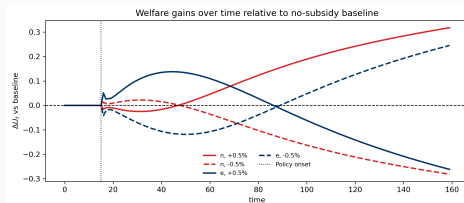
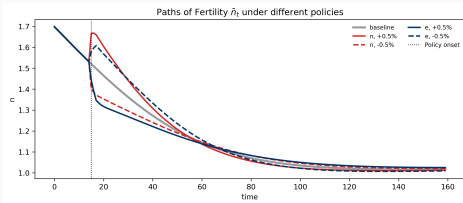
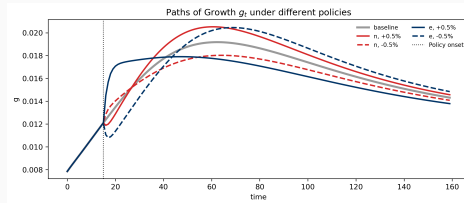
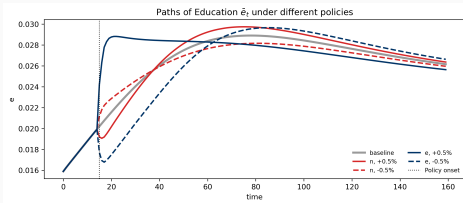
$$B_t = \sum_{i \in \{a, m, s\}} s_i(t) [s_e e_i(t) + s_n n_i(t)],$$

Actual consumption used in demand and welfare:

$$C_i(t) = \tilde{C}_i(t) - B_t.$$

The household does not internalize the effect that changing its decisions has on the lump-sum transfer $\partial B_t / \partial x_t = 0$ for all choice variables x_t .

Policy Counterfactuals: Paths



Policy Counterfactuals: CEV

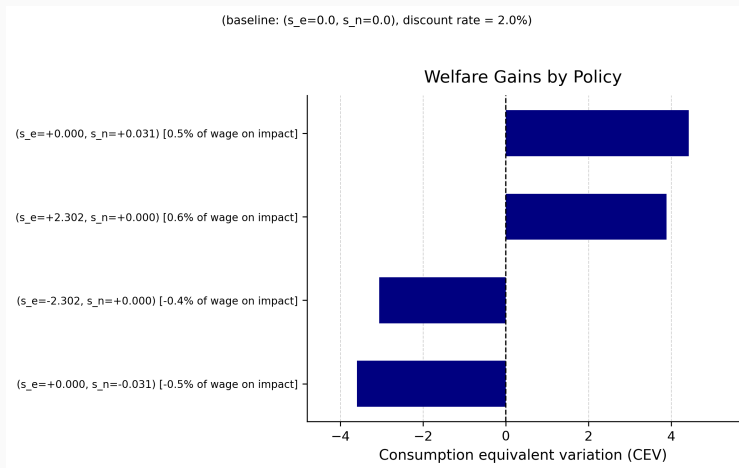


Figure 14: Ranking of policies by CEV

Conclusion

What the data show

- Higher peak manufacturing employment shares are associated with (i) faster GDPpc growth, (ii) higher education quality, and (iii) earlier fertility declines.

What the model does

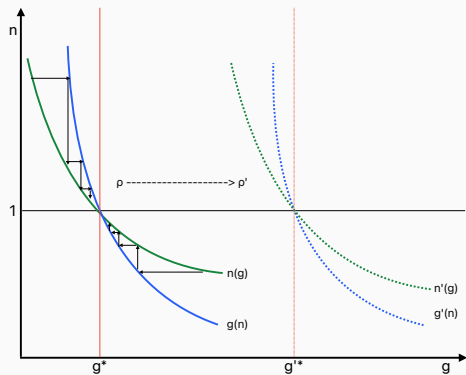
- Structural change affects development through a **demography–human capital** channel: sectoral employment composition shifts education incentives and fertility choices, shaping long-run income levels.

Policy implications

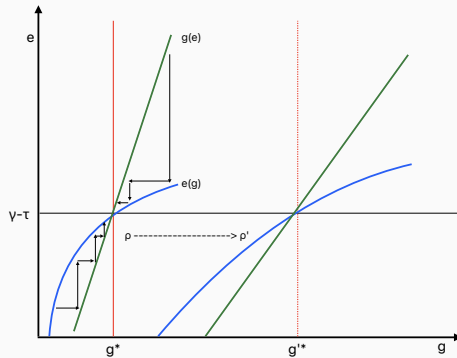
- **Trade policy:** openness reduces manufacturing employment share but welfare effects remain positive; gains depend on the opening date, comparative advantage and the ROW head-start.
- **Human capital policies:** education and fertility subsidies can have substantial CEV effects ($\approx 4\%$ for policies costing 0.5% of wage levels) but with different timings, fertility subsidies have gains which take a long time to materialize.

appendix

Model: Phase Diagram



(a) e and g



(b) n and g

Fact 1: Fertility and Structural Change

Table 2: OLS Regressions

	(1)	(2)	(3)	(4)	(5)
% Man. Emp.	-13.43*** (-10.74)	-4.815*** (-4.85)	-4.815*** (-4.85)	-6.113*** (-5.40)	-6.826*** (-5.54)
Log GDP p.c.		-0.808*** (-5.58)	-0.808*** (-5.58)	-0.782*** (-5.33)	0.355** (3.29)
% Agr.		0.809 (1.07)	0.809 (1.07)	0.559 (0.72)	0.0945 (0.12)
N	5053	5053	5053	5053	5053
R2	0.326	0.655	0.655	0.676	0.966
Country FE	no	no	no	no	yes
Year FE	no	no	no	yes	yes

t statistics in parentheses

Dependent variable is total fertility rate. Standard errors are clustered at the country level.

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Stylized Fact 1: Fertility and Structural Change

To quantify differences in fertility across sectors I run the following regression on a sample of 46 million women between the ages of 15 and 45 from 89 low and middle income countries:

$$nchild_{i,c,t} = \alpha_{a,c,t} + \delta_{e,c,t} + \gamma_{n,c,t} + \xi_{u,c,t} + \varepsilon_{i,c,t} \quad (5)$$

Where $nchild_{i,c,t}$ is the number of children for a survey respondent in country c and year t , α is a country-year-age fixed effect, δ is a country-year-education fixed effect, γ is a country-year-non-working fixed effect and ξ is a country-year-urban fixed effect.

Stylized Fact 1: Fertility and Structural Change

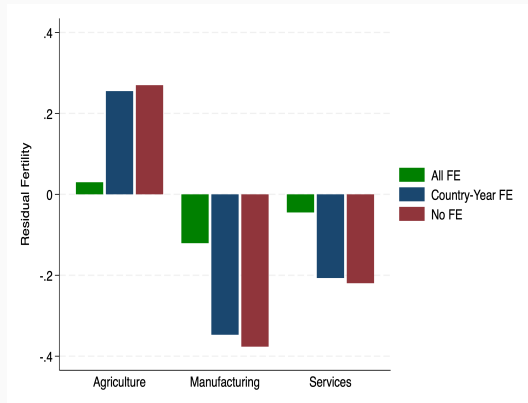


Figure 15: Average residual by sector from model 5 for different FE models. Data Source: IPUMS International.

Regression Table: GDP p.c. Growth rate in 2010-2020

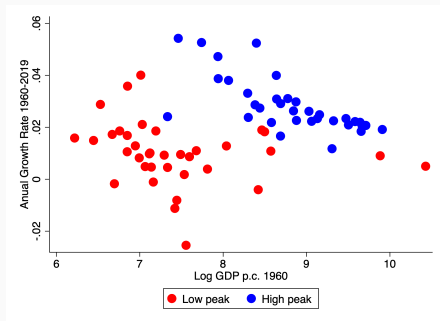
Table 3: Dep. Var. is Avg GDPPC growth rate 2010-2020

	(1)	(2)	(3)	(4)
Peak Emp. % Man.	8.517*** (4.07)	5.710** (2.38)	4.825* (1.71)	4.204 (1.63)
% Serv. Emp.	-4.124*** (-3.01)	-7.694*** (-3.10)	-7.433*** (-2.69)	-8.274*** (-2.96)
Log. GDP per capita		0.893** (2.12)	1.956* (1.78)	1.316 (1.07)
N	176	176	176	176
R2	0.0803	0.155	0.220	0.278
Income Decile FE	no	no	yes	yes
Region FE	no	no	yes	yes
WB Institutions	no	no	no	yes

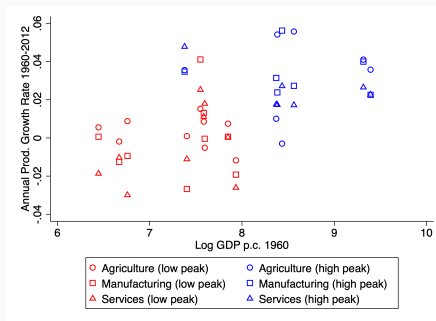
t statistics in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Stylized Fact 3: Lucas 2009 Style Plot



(a) Aggregate p.c. Growth



(b) Sector Decomposition

Figure 16: Scatterplot of average annual growth of GDP per capita over 1960-2020, distinguishing between high (blue) and low (red) peak manufacturing employment economies. Data source: Gapminder (left) and GGDC 10-sector (right).

Stylized Fact 3: Decomposition Across Sectors

Table 4: Decomposition of Prod. Gr. Effects

	(1)	(2)	(3)	(4)	(5)
	Agr.	Man.	Serv.	FIRE	Non-FIRE
Peak Emp. % Man.	-0.0129 (-0.29)	0.0699** (2.25)	0.0220 (0.49)	0.133** (2.60)	0.0140 (0.27)
N	41	41	41	41	41
R2	0.00358	0.108	0.00887	0.146	0.00313

t statistics in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Stylized Fact 2: Education Table 1/2

Table 5: Dependent Variable is avg. HCQ (1990-2020)

	(1)	(2)	(3)	(4)
Peak Emp. % Man.	6.512*** (6.34)	6.971*** (5.09)	2.055 (1.56)	1.990* (1.91)
% Man. Emp.		-1.461 (-0.74)	2.706* (1.83)	0.662 (0.43)
N	80	80	80	80
R2	0.303	0.308	0.569	0.695
Region FE	no	yes	no	yes
Controls	no	no	yes	yes

t statistics in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Stylized Fact 2: Education Table 2/2

Table 6: Dependent Variable is avg. TFR (1990-2020)

	(1)	(2)	(3)	(4)
HCQ (1990-2020)	-0.579*** (-8.06)	-0.449*** (-5.71)	-0.367*** (-4.27)	-0.216* (-1.91)
Peak Emp. % Man.		-2.810*** (-4.07)	-3.222*** (-4.31)	-2.962*** (-3.61)
N	80	80	80	80
R2	0.503	0.561	0.604	0.665
Region FE	no	no	no	yes
Controls	no	no	yes	yes

t statistics in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Stylized Fact 3: Heterogeneity in the Service Sector

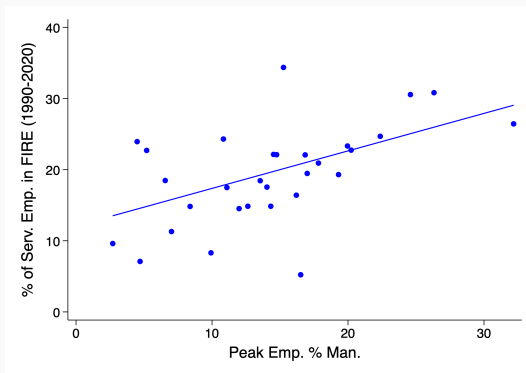
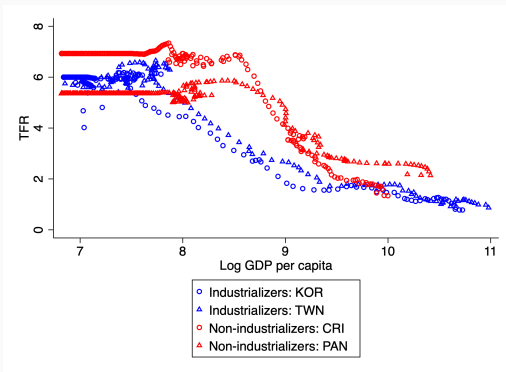
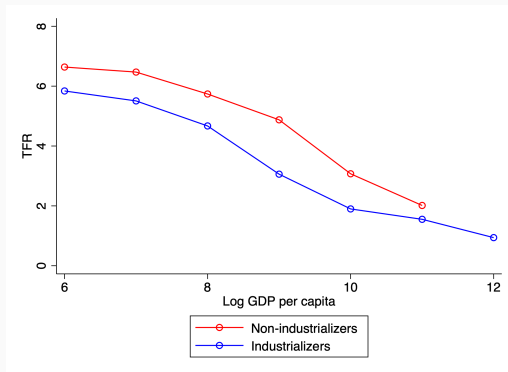


Figure 17: Binscatter plot of the fraction of service workers working in high productivity services (FIRE plus health-care and legal) against the peak employment share in manufacturing. Data source: IPUMS International for the fraction of high productivity services, GGDC for the peak employment shares of manufacturing.

Stylized Fact 1: Fertility and Structural Change



(a) Country-Comparison



(b) q1 vs q5

Figure 18: Scatterplot of the Total Fertility Rate (TFR) over log of GDP per capita in four countries. Red dots correspond to major industrializers and blue to non-industrializers. Data Source: Gapminder.

Competitive Equilibrium

A competitive equilibrium in this economy consists of household decisions c_a, c_m, c_s, n, e, i , firm decisions L_a, L_m, L_s , prices w_a, w_m, w_s, p_m, p_s , and sectoral technologies A_a, A_m, A_s . Such that, given prices:

- **Labor market:** Total labor supply equals total demand:

$$L = \sum_{i \in \{a, m, s\}} L_i.$$

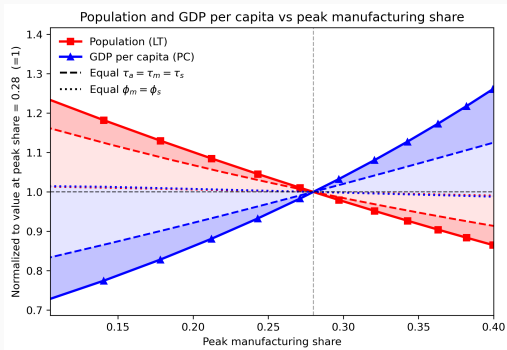
- **Goods markets:** Each sector's output equals demand:

$$y_i = c_i, \quad i \in \{a, m, s\}.$$

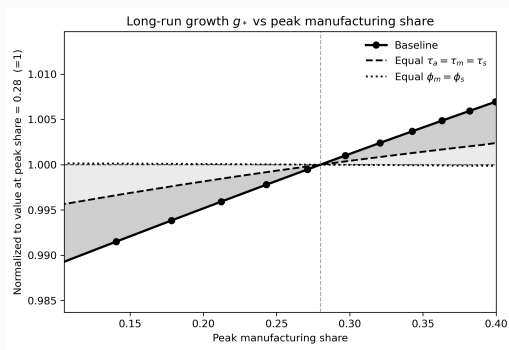
- **Optimization:** The household decisions solve the problem given in 1.

The economy is on a BGP if $g_{i,t} = g_{i,t+1} \forall i$.

Results: GDPPC and Population Decomposition



(a) $L_T, GDPPC_T$



(b) g^*

Figure 19: Final GDPPC and Population for economies with different peak employment shares in manufacturing.

Regression Table: GDP p.c. and Peak % Man. Emp.

Table 7: Dependent var is $\ln(\text{GDPPC})$

	(1)	(2)	(3)	(4)	(5)
Peak Emp. % Man.	9.308*** (11.80)	8.898*** (9.69)	3.182*** (3.91)	4.620*** (4.97)	1.950*** (2.63)
Years Since Peak		0.00595 (1.26)	-0.00136 (-0.36)	0.0103** (2.23)	0.00104 (0.28)
N	176	176	176	176	176
R2	0.426	0.430	0.766	0.605	0.812
Region FE	no	no	no	yes	yes
WB Institutions Controls	no	no	yes	no	yes

t statistics in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Fertility and Education Choices

Define $\tilde{w} \equiv \frac{\bar{c}}{1-\gamma}$, the level of w such that if the household allocates $(1 - \gamma)$ of its resources to consumption, it can just satisfy the subsistence constraint. Then for all households i :

$$n_{it}(\tau_i + e_{it}) = \begin{cases} \gamma & \text{if } w_t \geq \tilde{w}, \\ 1 - \frac{\bar{c}}{w_t} & \text{if } w_t \in (\bar{c}, \tilde{w}), \\ 0 & \text{if } w_t \leq \bar{c} \end{cases} \quad (6)$$

Implications for effective resources:

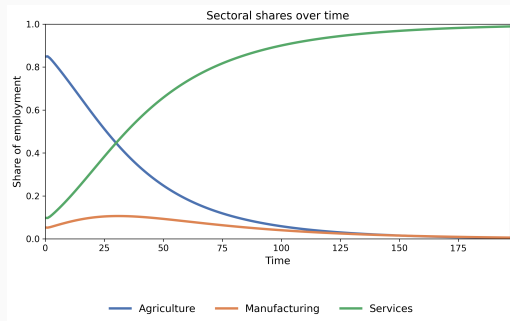
$$\underbrace{(1 - n_{it}(\tau_i + e_{it}))}_{R_t} w_t = \begin{cases} w_t(1 - \gamma) & \text{if } w_t \geq \tilde{w}, \\ \bar{c} & \text{if } w_t \in (\bar{c}, \tilde{w}) \end{cases}$$

Optimal education choice:

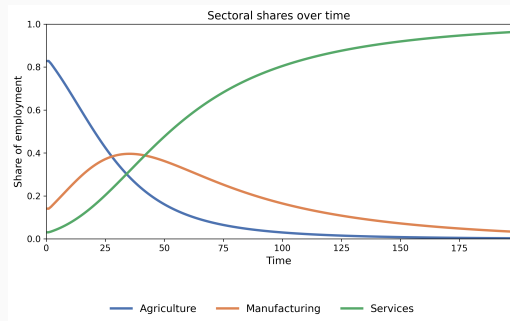
$$e_{it}^* = \max \left\{ 0, \sqrt{\bar{g}_t(1 - \rho_t)\tau_i} - \rho_t\tau_i \right\}$$

Model: Qualitative Patterns 1/2

Consider two economies with different ω_m :



(a) $\omega_m = 0.5$



(b) $\omega_m = 3.5$

Model: Qualitative Patterns 2/2

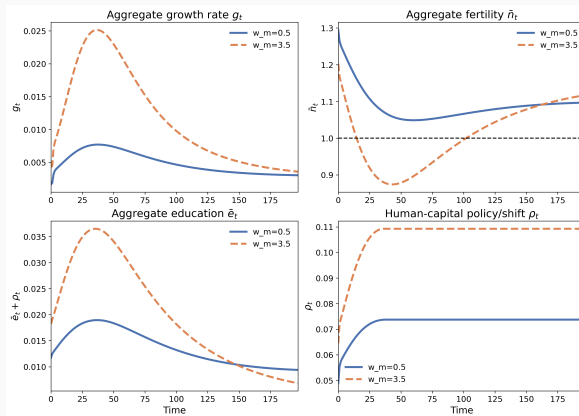
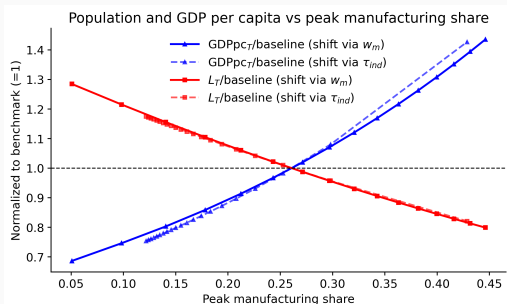
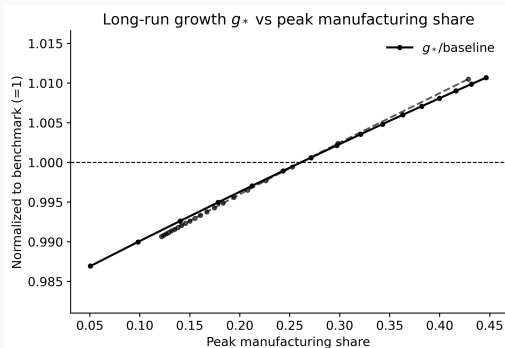


Figure 20: Comparison of Dynamics for two economies. The blue line corresponds to the economy which does NOT go through manufacturing (low peak), and the orange line the economy which does go through manufacturing (high peak).

Results: GDPPC and Population Policy Shifter vs ω_m



(a) $L_T, GDPPC_T$



(b) g^*

Figure 21: Final GDPPC and Population for economies with different peak employment shares in manufacturing. Comparison between counterfactuals obtained by changing the utility weight on manufactured goods ω_m , vs a policy wedge τ_{ind} which affects the relative price of services.

Table 8: Non-calibrated parameters (fixed and normalizations)

Parameter	Value	Source / normalization
ν	0.5	Cobb-Douglas (normalization)
σ	0.5	CES elasticity of substitution (standard)
ϵ_m	1	Manufacturing income-elasticity (normalization)
$\bar{r}$	0.05	Exogenous interest rate (normalization)
ω_m	3.5	Manufacturing utility weight (normalization)
T^{gen}	20	Generation length in years (Lagerlof, 2006)
h_0	1	Initial human capital (normalization)
A_0	[1.0, 1.0, 1.0]	Initial TFP levels (normalization)
$\bar{c}$	0.1	Subsistence / minimum consumption (fixed)

Residual Fertility by Sector: USA 1850-2020

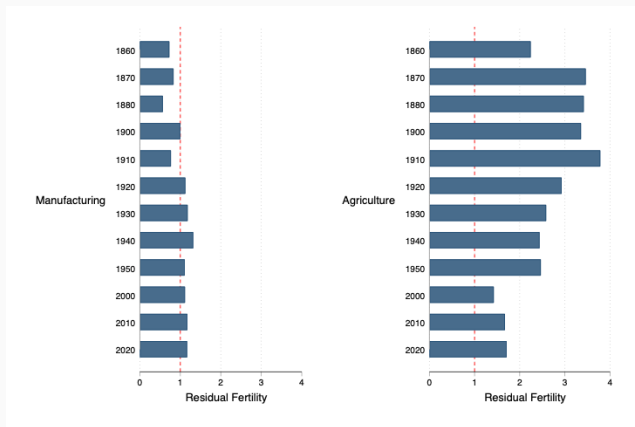


Figure 22: Sectoral fertility relative to services. Residuals from model 5 using samples of 25-35 year old women from the US census of population are computed and then normalized by the service sector fertility. For this exercise the model only includes a year fixed effect. Bars show the average residual in a year across sectors. Data source: IPUMS USA.

Table: Decomposing the Demographic Transition

Table 9: Total fertility decline (1850–2010): data and model counterfactuals

Scenario	Decline ($NFR_{1850} - NFR_{2010}$)	% of data decline
Data	1.453	100.00%
Model (no SC)	0.483	33.25%
Model (no education increase)	1.295	89.16%
Model (total)	1.548	106.57%

Calibration: GDP p.c.

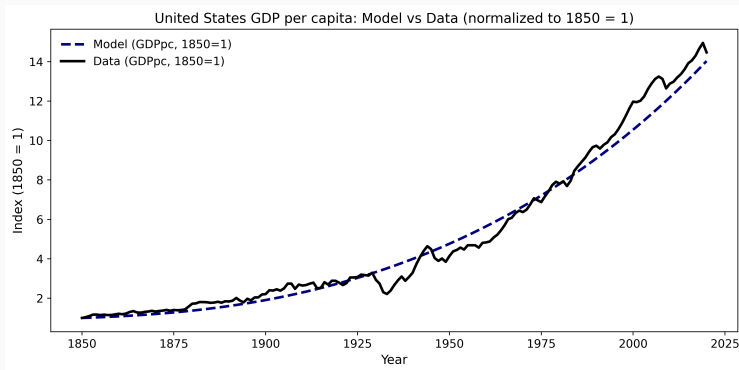


Figure 23: Calibration of GDP per capita over the development process. The solid black line shows data GDP per capita in the US over 1850-2020, the dashed blue line shows the model path. Data source: Maddison (2020).

Calibration: Relative Productivity

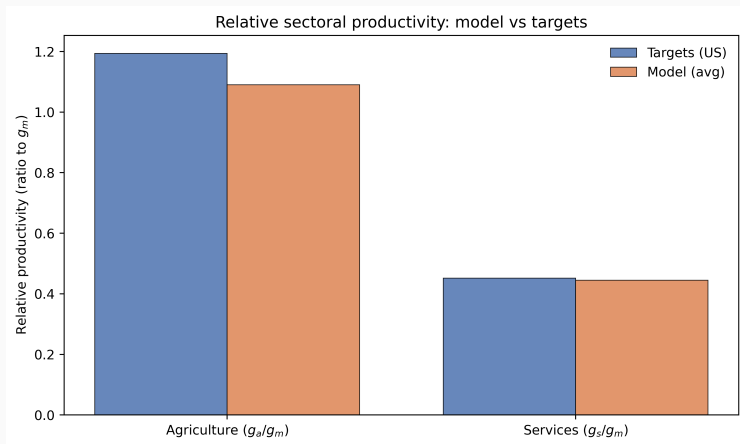


Figure 24: Relative productivity targets. Blue bars correspond to the relative productivities of agriculture and services computed using the GGDC 10-sector database, red bars show the equivalent in the model. Data source: GGDC 10-sector.

Calibration: Sectoral Fertility

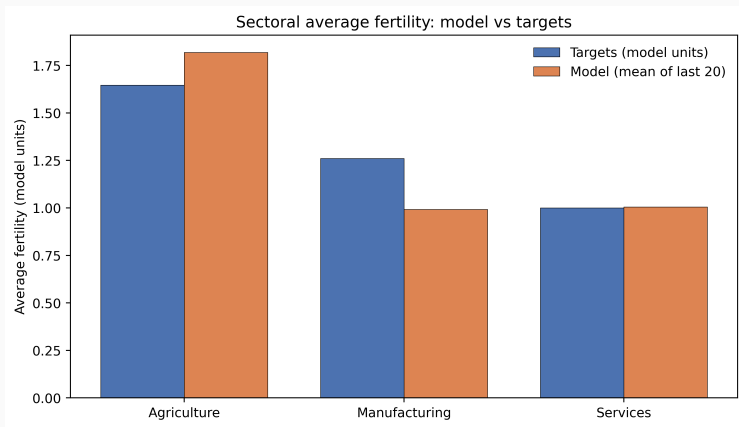


Figure 25: Sectoral Fertility: model vs targets. Blue bars correspond to the average fertility of agricultural, manufacturing and services households (assuming replacement fertility in services) in model units using the last two samples of IPUMS data. Red bars show the corresponding values in the model. Data source: IPUMS International.

CEV Formula

Letting W_t^{open} and W_t^{aut} denote per-period welfare under openness and protection, respectively, the CEV λ solves

$$\sum_{k=0}^{T-1} \beta^k (W_{t_0+k}^{\text{open}} - W_{t_0+k}^{\text{aut}}) = (1 - \gamma) \ln(1 + \lambda) \sum_{k=0}^{T-1} \beta^k,$$

where β is the discount factor per model period and $1 - \gamma$ is the weight on the consumption term in utility.

Solving for λ yields

$$\lambda = \exp\left(\frac{\sum_{k=0}^{T-1} \beta^k (W_{t_0+k}^{\text{open}} - W_{t_0+k}^{\text{aut}})}{(1 - \gamma) \sum_{k=0}^{T-1} \beta^k}\right) - 1.$$

CEV: Agriculture vs Manufacturing Comparative Advantage

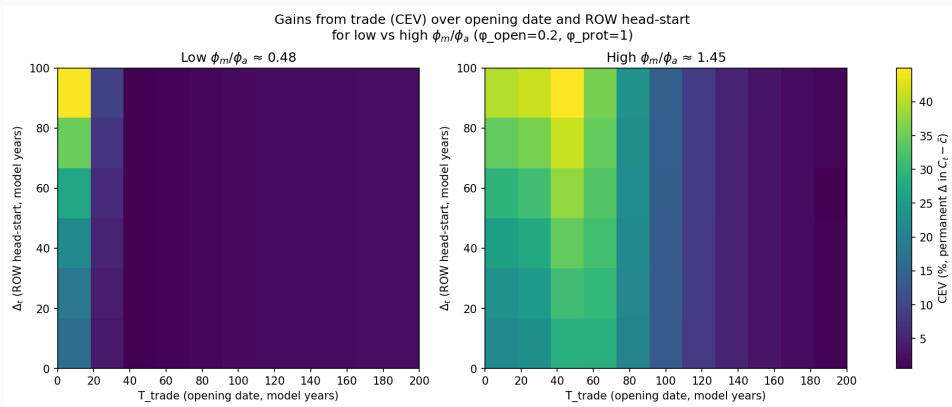


Figure 26: Welfare gains from trade openness over different opening dates and ROW head-start in two economies with different ϕ_m/ϕ_a ratios. The economy with a high ϕ_m/ϕ_a has a comparative advantage in manufacturing over ROW, where the ϕ_m/ϕ_a is ≈ 1 . The economy with a low ϕ_m/ϕ_a has a comparative advantage in agriculture.